PIONEER JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATIONS 2011

Candidate Name										
Civics Group						**INDEX NUMBER**				

H1 BIOLOGY **8875 / 01**

PAPER 1 Multiple Choice **23 September 2011**

Additional Materials: Multiple Choice Answer Sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

READ THESE INSTRUCTIONS FIRST

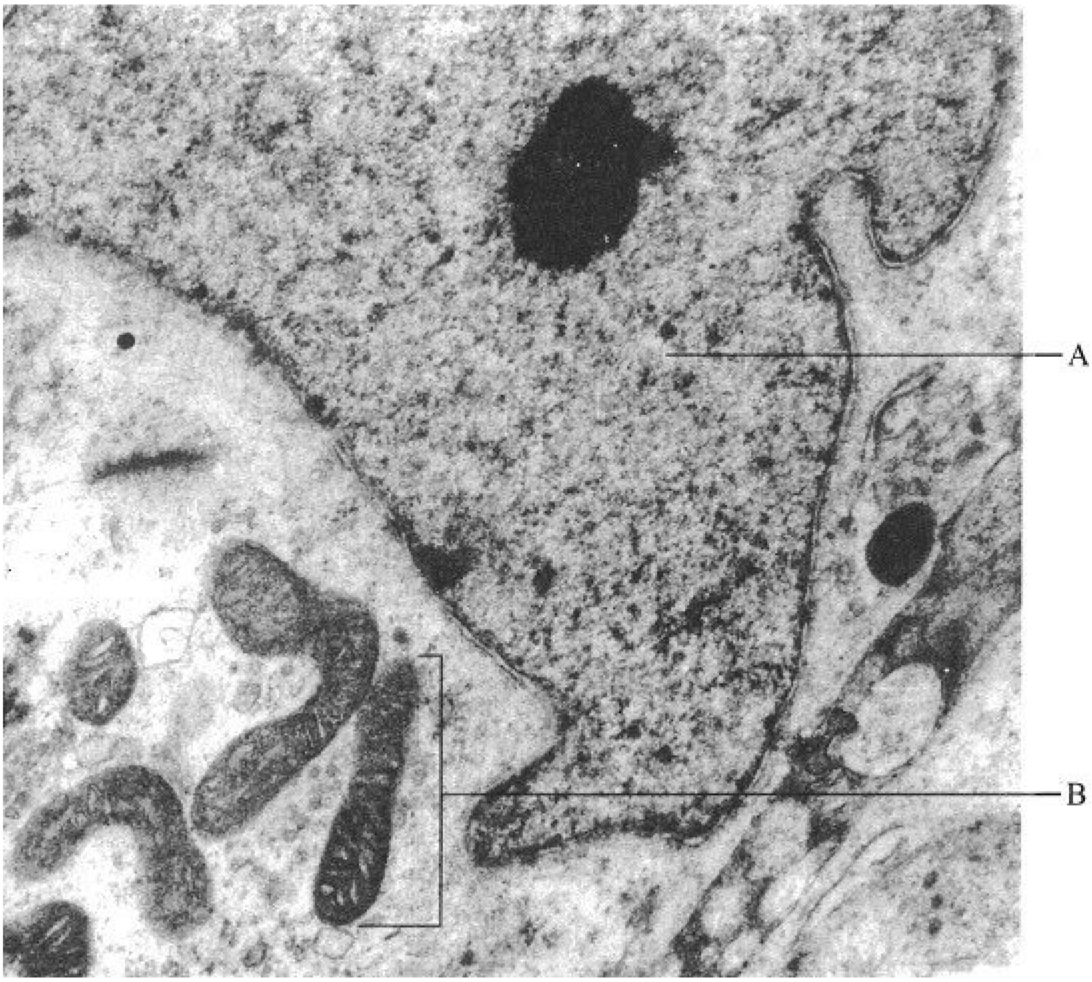
Write in soft pencil.
Do no use staples, paper clips, highlighters, glue or correction fluid.
Write your name, class and index number on the cover page in the spaces provided unless this has been done for you.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C and D**.
Choose the **one** you consider correct and record your choice **in soft pencil** on the OTAS answer sheet provided.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer. Any rough working should be done in this booklet.

Do not open this booklet until you are told to do so.

1. The photograph below-shows part of an animal cell, as seen using an electron microscope. The magnification is ×5000.



What are the functions of A and B and what is the actual length of B?

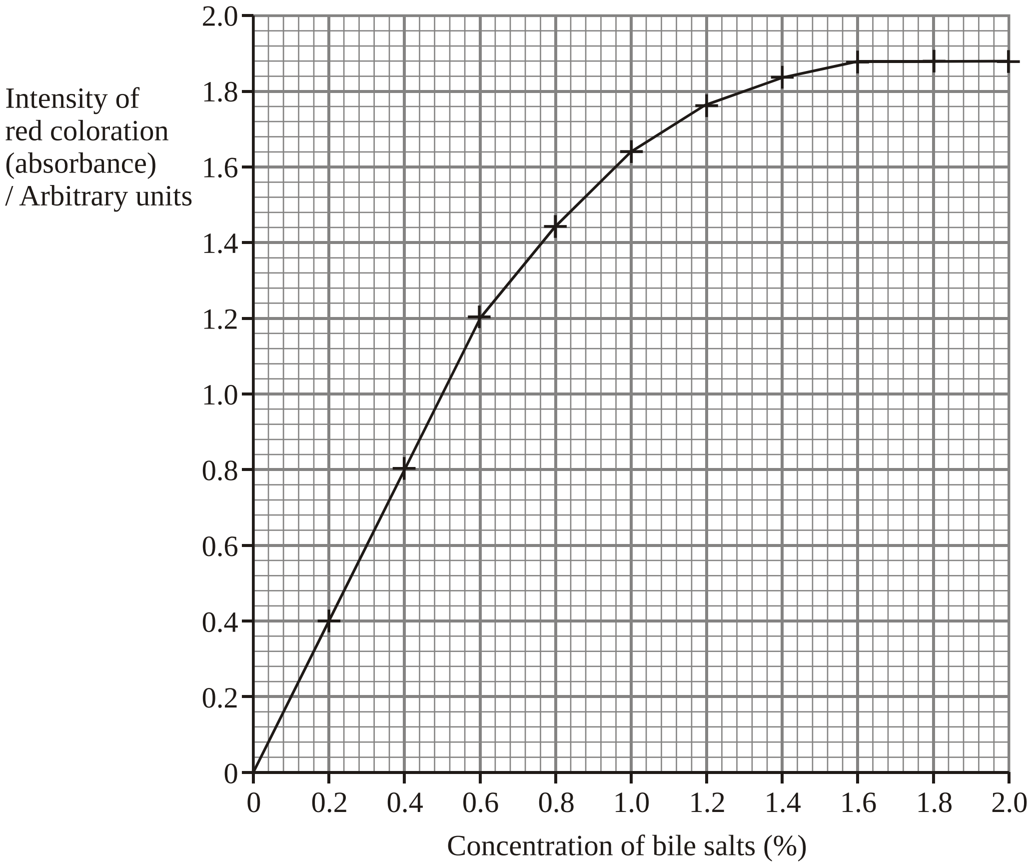
	A	B	Actual length of B
A	Regulates cell activity	Final chemical modification, sorting and packaging of proteins	72 µm
B	Maintains internal hydrostatic pressure within the cell	Post-translational modification	7.2 µm
C	Regulates cell activity	Site of ATP synthesis	7.2 µm
D	Maintains internal hydrostatic pressure within the cell	Site of lipid synthesis	72 µm

2. Beetroots are root vegetables. They appear red because their cells contain a water soluble red pigment in their vacuoles, which cannot pass through membranes.

Five beetroot discs were then placed in a tube containing 25 cm^3 of 2% bile salt solution and left for 30 minutes at 20 °C. The procedure was repeated for different concentrations of bile salts and one set of discs was left in distilled water.

After 30 minutes, each set of beetroot discs was removed from the solutions and from the water. Each bile salt solution was stirred and a sample removed and placed in a colorimeter. The intensity of red coloration (absorbance) was determined by the colorimeter.

The results of the investigation are shown in the graph below.



What is the best explanation for the results?

A Pigments leaked out via protein channels in the plasma membrane.

B Bile salts caused phospholipids in both tonoplast and plasma membrane to be emulsified.

C Bile salts did not have any effect on the tonoplast.

D Temperature affected the intensity of colour absorbance.

3. A sample of living cells from the pancreas is supplied with amino acids containing the heavy isotope of nitrogen, ^{15}N.

   Which structures in the cells from the pancreas would show the highest concentration of ^{15}N?

A centrioles and nuclear envelope

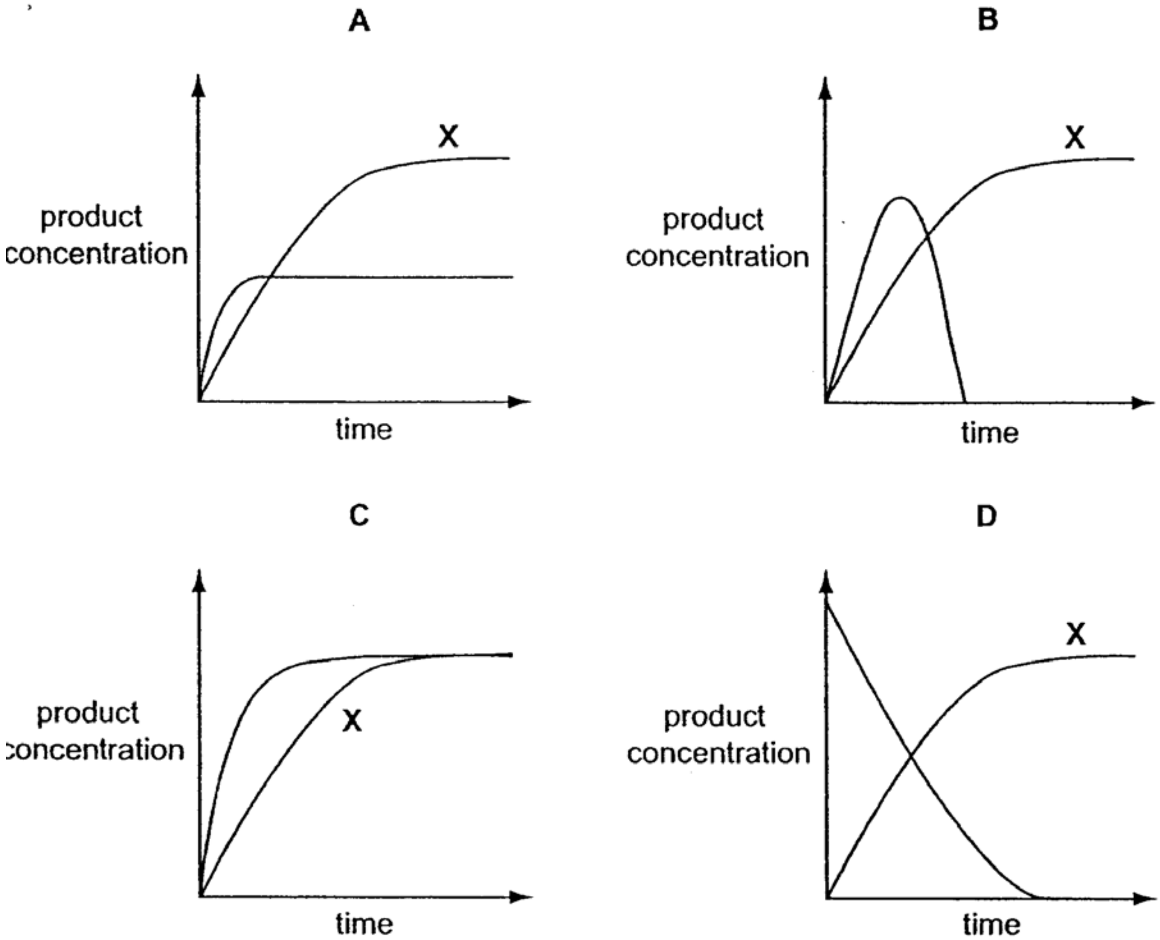
B endoplasmic reticulum and mitochondria

C lysosomes and nucleus

D ribosomes and Golgi body

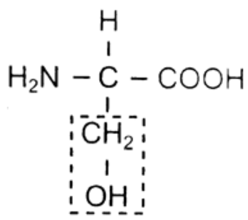
4. Two enzyme experiments were carried out. The first, experiment **X**, was carried out at a constant temperature of 37°C. During the second experiment the temperature was increased from 37°C to 80°C.

   Which graph shows the results?

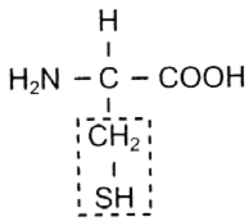


5. The diagram shows four different amino acids, each with a different R group (side chain).

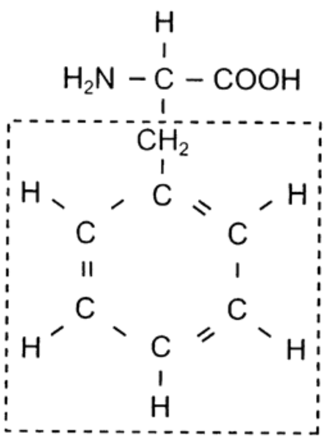
1



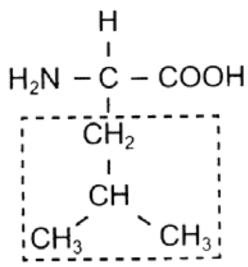
2



3



4



Which amino acids could form a hydrophobic interaction between R groups?

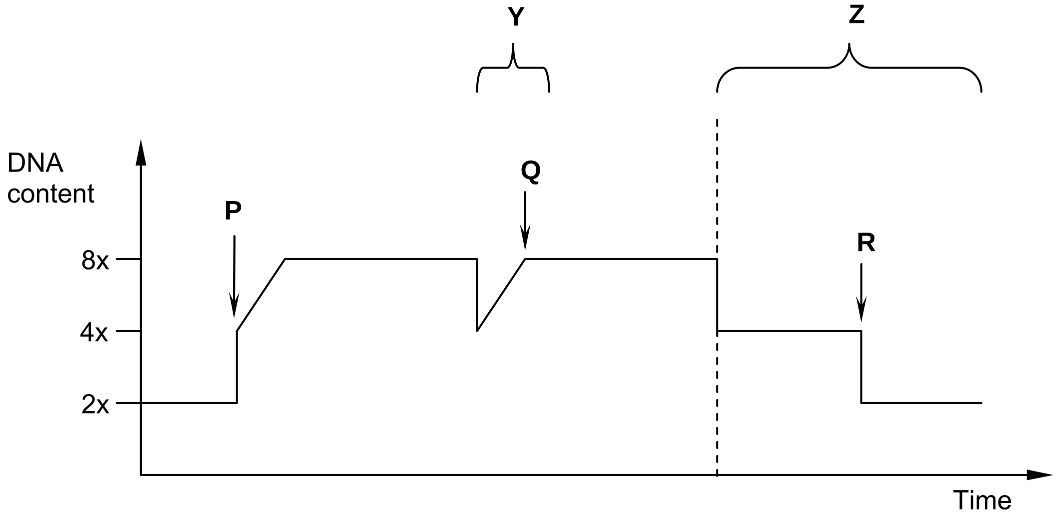
A 1 and 2

B 1 and 3

C 2 and 4

D 3 and 4

6. The graph represents the changes in the DNA content within a cell at different stages in the cell cycle.



Name the events occurring at **P**, **Q** and **R**, and identify the stage where meiosis was occurring.

	P	Q	R	Meiosis occurs at
A	S phase	Fertilisation	Cytokinesis	**Y**
B	Fertilisation	Interphase	Cytokinesis	**Z**
C	S phase	Prophase	Telophase	**Y**
D	Fertilisation	Metaphase	Cytokinesis	**Z**

7. The table below shows some amino acids and their DNA genetic code.

Amino acid	**DNA code**
Alanine	CGG
Lysine	TTT
Arginine	GCG
Phenylalanine	AAA
Glycine	CCA
Valine	CAA

A DNA sequence codes for a polypeptide arginine-glycine-phenylalanine-valine-alanine.

The following mRNA molecule was transcribed from the DNA sequence above. Which mRNA triplet contains a single transcription error?

CGCGUUUUUGUUGCC

A the first

B the second

C the third

D the fourth

8. Which of the following shows the possible effects of a single nucleotide substitution in each of the following locations in a gene on the production of the protein product of that gene?

	Promoter	Terminator	Start codon	Stop codon	Middle of an intron
A	No protein product is produced	Protein product is shorter than normal	Protein product is longer than normal	Protein product is normal	Too much protein product is produced
B	Too much protein product is produced	Protein product is normal	No protein product is produced	Protein product is longer than normal	Protein product is normal
C	Protein product is normal	Protein product is longer than normal	Protein product is shorter than normal	Too much protein product is produced	Protein product is longer than normal
D	Protein product is longer than normal	Too much protein product is produced	Protein product is normal	Protein product is shorter than normal	No protein product is produced

9. Five different amino acids (numbered **1** to **5** below) form the following sequence in part of a polypeptide chain:

2------3------4------1------3------5------2

Messenger RNA (mRNA) codons which correspond to these amino acids are:

amino acid **1**	UGU
amino acid **2**	GAU
amino acid **3**	CAC
amino acid **4**	CAA
amino acid **5**	AAG

Which one of the following DNA base sequences could provide the code for the given section of polypeptide?

A CTAGTGGTTACAGTGTTCCTA

B CTAGTGGTTACAGTGTTGCTA

C CTAGTGGTTTCTGTGTTGCTA

D CTAGTGGTTACAGTGTTCCTT

10. Genetic drugs are short sequences of nucleotides. One example is an anti-sense drug made from RNA nucleotides that are complementary to the protein-coding mRNA.

Which is the process that is affected by the anti-sense drug and what is its most probable mode of action?

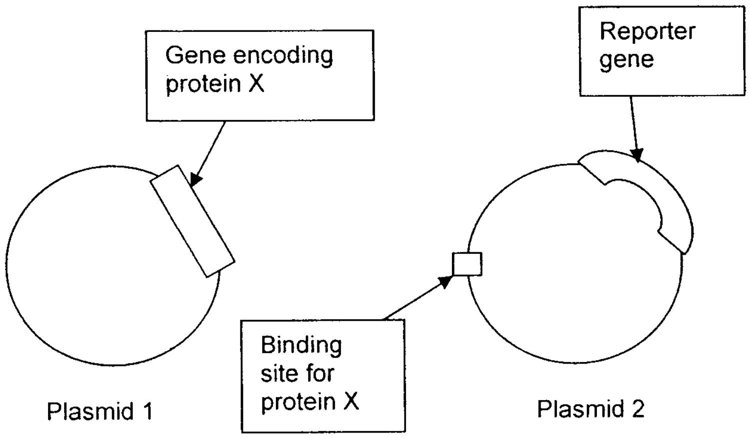
A It prevents DNA replication by binding to the anti-sense strand of DNA that codes for the disease-causing protein.

B It prevents transcription by binding to the sense strand of DNA that codes for the disease-causing protein.

C It prevents translation by binding to the mRNA that contains information for the disease-causing protein.

D It prevents the normal functioning of disease-related protein when it is translated into a protein that inhibits the disease-causing protein.

11. An investigator wanted to determine the function of protein X. He cloned the gene X into plasmid 1. He then set up another plasmid (plasmid 2) with a binding site for the product of gene X plus a reporter gene. A reporter gene codes for the synthesis of a product that is easily measured. The two plasmids (shown below) were then introduced into the nucleus of a host cell.



He noted that there was an increase in the number of reporter gene transcripts. Which of the following is a valid conclusion regarding gene X or product Y?

A Gene X is an origin of replication sequence.

B Gene X is a promoter sequence.

C Product X is DNA polymerase.

D Product X is a transcription factor.

12. Which of the following statements about cancer is / are true?

I	Cancer is a result of increased cell division which promotes the mutation of a proto-oncogene.
II	Individuals who inherit one inactive copy of tumour suppressor gene are more likely to develop cancer than individuals with two non-mutant copies.
III	Mutagenic activation of a single oncogene is sufficient to convert a normal cell into a cancer cell.

A II only

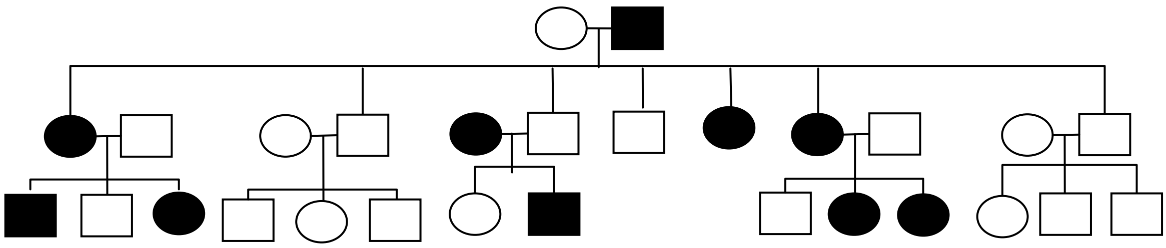
B I and III only

C II and III only

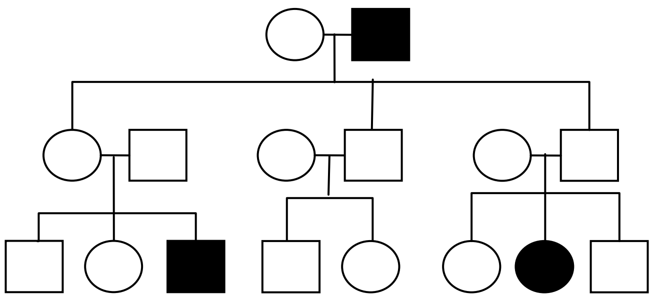
D All of the above

13. In 2003, scientists discovered two new genetically transmitted diseases, **P** and **Q**, commonly found among a community living in the South of France. The pedigree diagrams below show the inheritance of these diseases. A general key to these diagrams is also given.

Disease P



Disease Q



Key:

The alleles responsible for causing disease **P** and **Q** respectively are as follows:

A autosomal dominant, sex-linked recessive

B autosomal recessive, sex-linked dominant

C sex-linked dominant, autosomal recessive

D sex-linked recessive, autosomal dominant

14. In mice, the gene for ‘dappled’ coat (D) and its recessive allele, the gene for ‘plain’ coat (d), are located on the X chromosome. The gene for ‘straight’ whiskers (W) and its recessive allele, the gene for ‘bent’ whiskers (w), are autosomal. A male mouse with plain coat and bent whiskers was mated on several occasions to the same female and the large number of offspring consisted of the following phenotypes in equal proportion:

dappled male with straight whiskers	plain male with straight whiskers
dappled female with straight whiskers	plain female with straight whiskers
dappled male with bent whiskers	plain male with bent whiskers
dappled female with bent whiskers	plain female with bent whiskers

If X^D represents an X chromosome carrying a gene for ‘dappled’ coat and X^d represents an X chromosome carrying a gene for ‘plain’ coat, what is the genotype of the female parent?

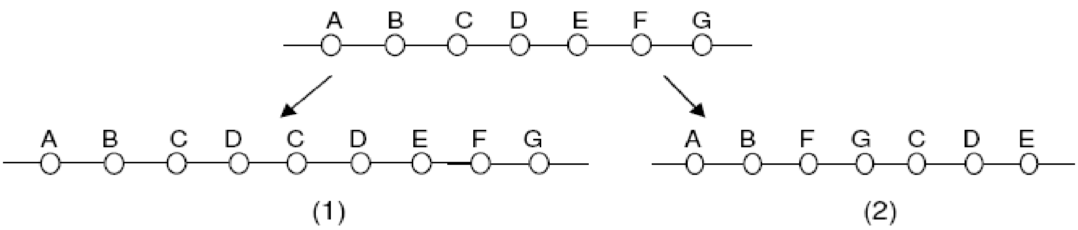
A $X^D X^D WW$

B $X^D X^D Ww$

C $X^D X^d WW$

D $X^D X^d Ww$

15. The diagrams below show two types of chromosomal mutation producing changes from the normal gene sequence.



Which of the following shows the two forms of chromosome mutation shown above?

	(1)	(2)
A	Deletion	Duplication
B	Duplication	Translocation
C	Duplication	Inversion
D	Inversion	Duplication

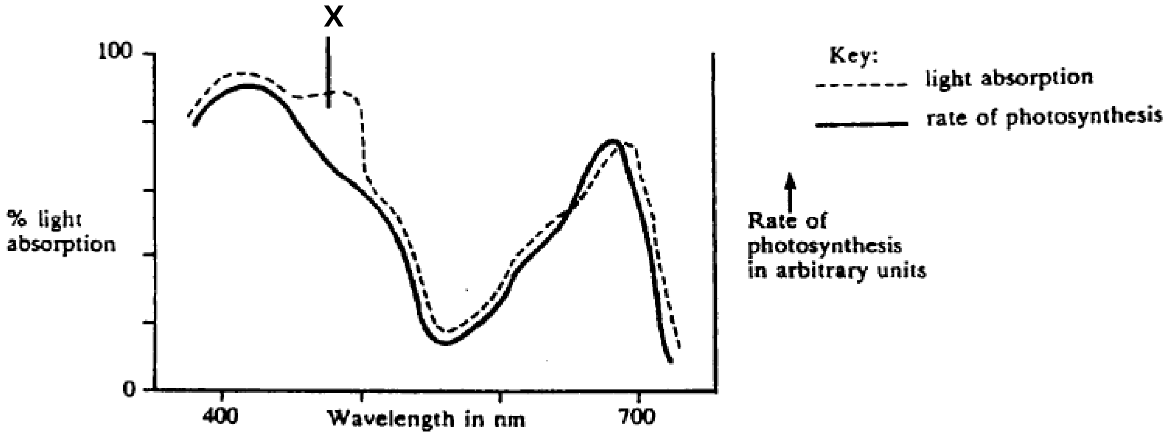
16. The Himalayan variety of rabbits has white hair on the body and black hair on the feet, tail, ears and face. The allele for the Himalayan rabbit pigment pattern, c^h, is recessive to the alleles for normal colour (all hair agouti), C, as well as dark chinchilla (all hair dark grey), c^{chd}, and is dominant to the allele for albino (all hair white, no pigment production), c. All of the alleles of this gene produce different versions of the same enzyme involved in pigment production.

A patch of white fur was removed from a Himalayan rabbit and an ice pack secured to the skin. The fur that grew back on the patch was black.

Which is correct?

	Genotypes of Himalayan rabbits	**Explanation for pigment pattern in Himalayan rabbits**
A	c^hc^h only	the enzyme is denatured at the high skin temperatures found on the rabbits' bodies
B	c^hc^h only	the enzyme becomes inactive at the low skin temperatures found on the rabbits' feet, tail, ears and face
C	c^hc^h and c^hc only	the enzyme is denatured at the high skin temperatures found on the rabbits' bodies
D	c^hc^h and c^hc only	the enzyme becomes inactive at the low skin temperatures found on the rabbits' feet, tail, ears and face

17. The graph below shows the effect of different wavelengths of light on the rate of photosynthesis and on the amount of light absorbed by the pigments in a green seaweed.



The difference between the two curves at X is due to

A inefficient trapping of light energy by the chlorophyll

B no ATP production at that wavelength

C oxygen given off during photosynthesis interferes with the absorption of light

D carotenes absorbing light that is not used in photosynthesis

18. The concentration of carbon dioxide in a sample of air was found to be 280 ppm (parts per million). A controlled experiment was designed to measure the concentration of carbon dioxide in the air after it had flowed over the leaves of a green plant. Measurements were taken at a range of light intensities.

The following results were obtained:

Light intensity (% of full sunlight)	Concentration of carbon dioxide in air after flowing over leaves (ppm)
75	253
50	252
25	254
10	280

Which one of the following statements is not consistent with these results?

A At the lower light intensities tested, the rate of photosynthesis is limited by light intensity.

B At the higher light intensities tested, the rate of photosynthesis is affected by factors other than light.

C In the dark, the concentration of carbon dioxide in the air after it had flowed over leaves would be at least 280 ppm.

D At a light intensity of 10% of full sunlight, photosynthesis does not occur.

19. Protons are pumped into the intermembrane space in the mitochondrion during electron transport. These protons diffuse through proteins to be used in the synthesis of ATP.

Why do these protons diffuse only through the proteins in the membrane?

A They are repelled by the charge on the phospholipids.

B They are too large to pass between the phospholipids.

C They are water soluble and cannot pass through the phospholipids.

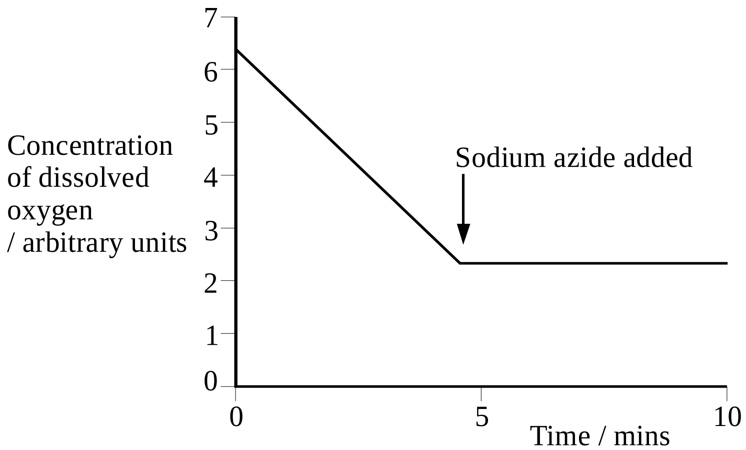
D They attach to the polar heads of the phospholipids.

.

20. Active mitochondria can be isolated from liver cells. If these mitochondria are then incubated in a buffer solution containing a substrate, such as succinate, dissolved oxygen will be used by the mitochondria. The concentration of dissolved oxygen in the buffer solution can be measured using an electrode.

An experiment was carried out in which a suspension of active mitochondria was incubated in a buffer solution containing succinate, an intermediate of the Krebs cycle. The concentration of dissolved oxygen was measured every minute for five minutes. A solution containing sodium azide was then added to this preparation and the concentration of dissolved oxygen was measured for a further five minutes.

The results are shown in the graph below.



Which of the following statements does not explain the results shown in the graph?

A The decrease in the concentration of dissolved oxygen in the first 5 minutes is due to oxygen combining with hydrogen ions and electrons to form water.

B Sodium azide inhibits cytochrome oxidase from binding to water.

C When sodium azide is added, the electron flow down the electron transport chain is inhibited.

D Succinate cannot be oxidised when sodium azide is added.

21. The enzyme amylase hydrolyses starch into disaccharides. Amylase is unable to breakdown cellulose because

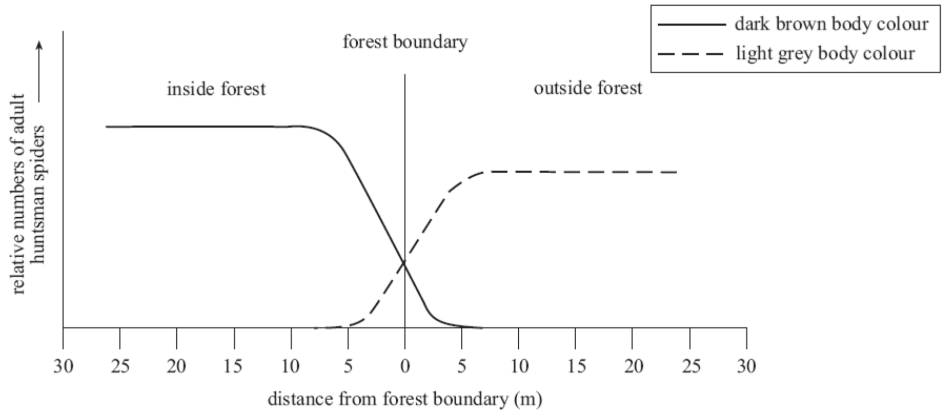
A cellulose does not contain glycosidic bonds.

B starch consists of glucose; cellulose consists of other monosaccharides.

C the bonds between sugars in cellulose are much stronger.

D the sugars in cellulose are bonded together differently than in starch, hence cellulose has a different shape.

22. The graph below shows the distribution of huntsman spiders at a forest boundary.



One species of huntsman spider (*Isopeda isopedella*) varies in body colour from dark brown to light grey. In one community at the forest boundary, two populations of this species were found. Some were found living in the leaf litter inside the forest and others were found living in the grass just outside the forest. The relative numbers of dark brown adult spiders and light grey adult spiders found at certain distances from the forest boundary are shown in the graph above.

The best explanation for this distribution is that

A the two populations of spiders were once different species.

B the two populations of spiders were unable to interbreed and individuals were adapting to suit their habitats.

C the differences in the two habitats had changed the physical appearance of individual spiders.

D a particular body colour provided a selective advantage to spiders in a particular habitat.

23. During the drought years on the island of Daphne Major, small seeds became rare, leaving behind only large seeds as a food source for birds. In the event of a prolonged drought lasting several years, which of the following could be a possible observation?

A Large-beaked birds eating less, leaving food available for the small-beaked birds.

B Small-beaked birds developed large beaks to feed on the large seeds.

C Small-beaked birds dying, with subsequent generations having a higher percentage of large-beaked birds.

D Small-beaked birds mutating their genes to grow large beaks.

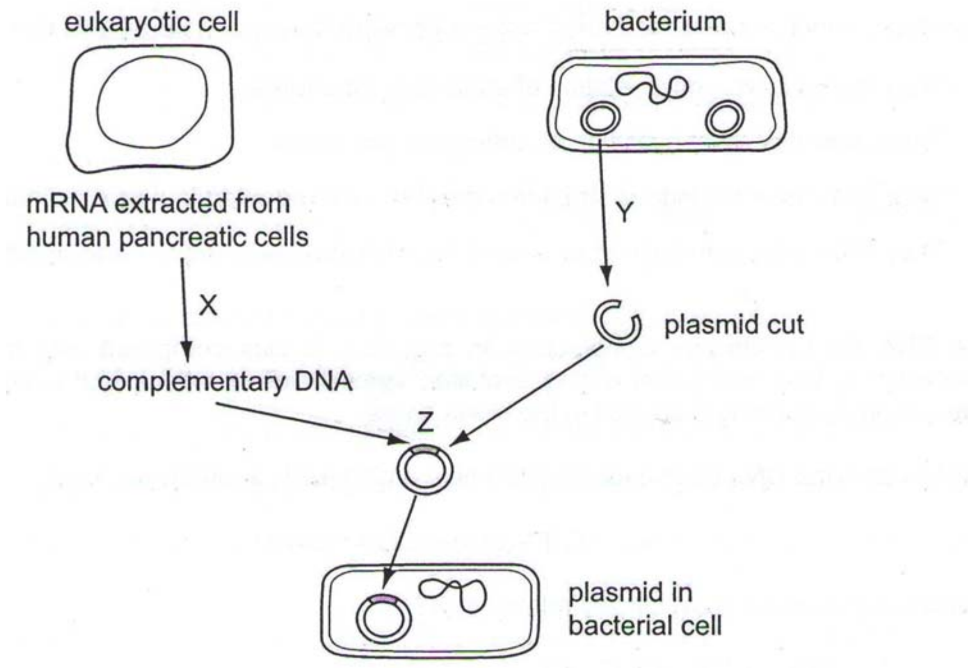
24. The table shows equivalent amino acid sequences of part of a protein from four species of animal.

animal	amino acid sequence							
1	trp	met	val	glu	cys	asp	arg	leu
2	trp	val	met	glu	cys	asp	asp	ala
3	trp	val	val	glu	cys	asp	arg	leu
4	phe	trp	val	gly	cys	arg	asp	leu

Using the technique of molecular homology, which pair of animals share the most recent common ancestor and which pair are least closely related?

	Most closely related pair	**Least closely related pair**
A	1 and 2	1 and 4
B	1 and 3	1 and 4
C	1 and 3	2 and 4
D	2 and 3	3 and 4

25. The diagram represents stages in producing genetically modified bacteria.



Which enzymes are used for processes X, Y and Z?

	X	Y	Z
A	DNA polymerase	Ligase	Restriction endonuclease
B	restriction endonuclease	DNA polymerase	ligase
C	reverse transcriptase	restriction endonuclease	DNA polymerase
D	reverse transcriptase	restriction endonuclease	ligase

26. Some of the goals of the Human Genome Project are listed.

1 to determine the base sequence of the whole of the human genome

2 to identify the genes encoded by the base sequence

3 to find the locations of the genes on the 23 pairs of human chromosomes

Which goals had to be achieved to allow the development of a single-stranded DNA probe for detecting the presence of a mutant allele on chromosome 7?

	1	2	3
A	✓	✓	✓
B	✓	✓	
C	✓		✓
D		✓	✓

27. It has been found that stem cells transferred from the intestinal lining to the bone marrow produce all of the different types of blood cell instead of intestinal cells.

Which statement explains this?

A All stem cells are totipotent.

B Environmental factors change the expression of specific genes.

C Specific genes are destroyed by endonucleases.

D Specific genes are hidden by condensation of some chromosomes.

28. One type of genetically modified corn has a gene for the production of *Bt* toxin, which protects the plant against a specific insect, as well as an 'amp' gene which was introduced into the plant together with the *Bt* toxin gene. The 'amp' gene gives resistance to the antibiotic ampicilin.

There is concern that the 'amp' gene may be transferred to enterobacteria in the human intestine during nucleic acid digestion, making treatment with ampicilin ineffective for diseases caused by enterobacteria.

Which statement explains why the transfer of this gene from the plant to bacteria in the human intestine is unlikely?

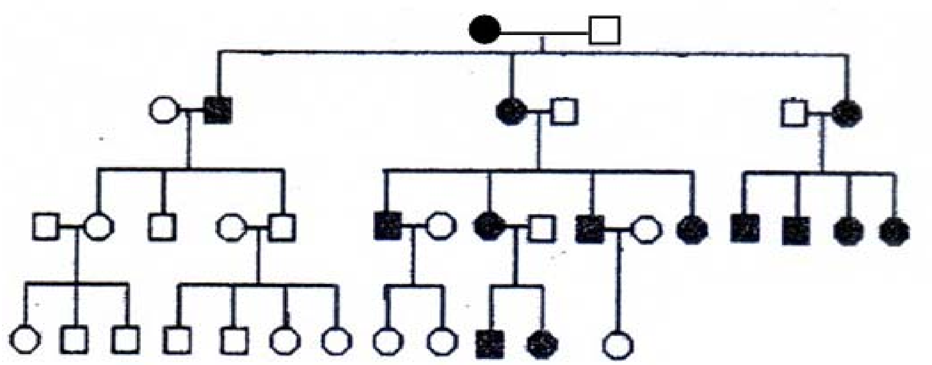
A An origin of replication and appropriate prokaryotic promoters are required for the 'amp' gene to be expressed.

B Bacteria took up DNA released during digestion of the plants by human nuclease enzymes, without the presence of a vector.

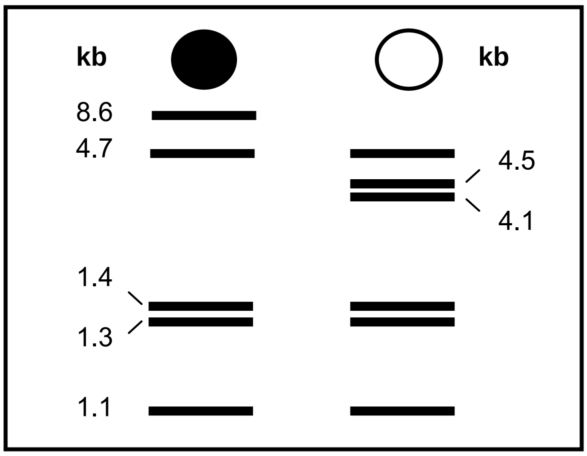
C 50% of the enterobacteria isolated from humans are 'amp' resistant.

D All plant DNA is digested and destroyed in the human intestine during digestion of plant cells by human nuclease enzymes.

29. The pedigree shows the inheritance of an RFLP marker locus which contains restriction sites for *Hae*II in mitochondrial DNA (mtDNA). mtDNA exists as a double-stranded circular DNA molecule, very much resembling a bacterial plasmid. The filled symbols indicate individuals who have lost a *Hae*II site to a point mutation.



mtDNA from an affected individual and an unaffected individual is digested with *Hae*II to produce restriction fragments that are separated using gel electrophoresis. The results are shown as follows:



Which of the following statements is not correct?

A There were six *Hae*II restriction sites in the mtDNA before the mutation.

B Affected females always transmit the trait to all their children.

C Mitochondrial genes in males are not transmitted to any of their children.

D The mutation occurred at the restriction sites between fragments of 4.1 kb and 4.7 kb.

30. Developing fish eggs can be treated to produce a diploid egg. In salmon, such eggs have been fused with haploid salmon sperm to give infertile triploid salmon. Reproductive organ tissue from diploid trout was transplanted into newly hatched triploid salmon. This tissue matured as the fish grew and the salmon successfully produced viable trout sperm or eggs which resulted in young trout.

Which fish could be seen as genetically identically modified?

A salmon providing eggs for treatment

B trout providing reproductive organ tissue

C young triploid salmon

D young trout

~THE END~